

CIAN ROCHE

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EDUCATION

Massachusetts Institute of Technology 2021 – Present

Ph.D. Candidate, Physics

Thesis: “*The Dark Universe Through The Strong Lens: Cluster and Galaxy-Scale Probes of the Dark Sector*”

Advisors: Prof. Michael McDonald, Prof. Mark Vogelsberger

Universität Tübingen 2019 – 2021

M.Sc., Mathematical Physics, 1.1 GPA[†]

Thesis: “*Exact parallel waves in general relativity*”

Advisor: Prof. Carla Cederbaum

University College Cork 2015 – 2019

B.Sc., Physics, 1st class honours

HONORS AND AWARDS

MIT Physics Henry W Kendall Teaching Fellowship	2026
DAAD Graduate Program Full Scholarship	2019
University College Cork College Scholar	2019
Max Planck Institute for Plasma Physics Research Scholarship	2018
Naughton Research Fellowship (one of 8 in Ireland)	2017
Naughton Foundation Scholarship (one of 29 in Ireland)	2015
All-Ireland Scholarship (one of 125 in Ireland; declined)	2015

PUBLICATIONS

ADS library: <https://tinyurl.com/CianRocheADS>

57 first-author citations, 129 citations, *h*-index 6

First Author or Significant Contribution

1. **Roche, C.**, Aazami, A.B. & Cederbaum, C. 2023; “*Exact parallel waves in general relativity*” *Gen Relativ Gravit* 55, 40, [doi:10.1007/s10714-023-03083-x](https://doi.org/10.1007/s10714-023-03083-x)
2. **Roche, C.**, Necib, L., Lin, T., Ou, X., Nguyen, T. 2024; “*The Escape Velocity Profile of the Milky Way from Gaia DR3*”, *ApJ*, 972, 70, [arXiv:2402.00108](https://arxiv.org/abs/2402.00108)
3. **Roche, C.**, McDonald M., Vogelsberger, M., Shen, X., Springel, V., Hernquist, L., Pakmor, R., Bose, S., Kannan, R. 2024; “*Brightest Cluster Galaxy Offsets in Cold Dark Matter*”, *OJA*, astro-ph.GA 7, [arXiv:2402.00928](https://arxiv.org/abs/2402.00928)
4. Tran, V., Shen, X., Vogelsberger, M., Gilman, D., O’Niell, S., **Roche, C.**, Zier, O., Gao, J. 2024; “*Novel Density Profile for Isothermal Cores of Dark Matter Halos*”, *Phys. Rev. D* 113, 063024, [arXiv:2411.11945](https://arxiv.org/abs/2411.11945)
5. **Roche, C.**, Vogelsberger, M., McDonald, M., Dutra, I., Natarajan, P., Shen, X., Shreeram, X., Metcalf, R., Sharon, K., Birrer, S., Lee, W., Meneghetti, M. 2026; “*A Consistent Implementation of Cluster Strong Lensing in Cosmological Simulation Light Cones*”, *OJA*, astro-ph.GA 7, [arXiv:2605.30433](https://arxiv.org/abs/2605.30433)
6. Fosdick, K., McDonald, M., Khullar, G., Gladders, M., Häußler, B., **Roche, C.**, et al. 2026; “*Quantifying the Dominance and Emergence of Brightest Cluster Galaxies In SPT and Planck Clusters: A Novel Application of the Gini Index*”, *accepted, ApJ*
7. **Roche, C.**, McDonald, M., Natarajan, P., Sharon, K., Dutra, I., Meneghetti, M., Vogelsberger, M. 2026; “*Host-Dependence and Line Of Sight Effects on Galaxy-Galaxy Strong Lensing in Clusters*”, *submitted, ApJ*, [arxiv:2608.30079](https://arxiv.org/abs/2608.30079)
8. Dutra, I., Natarajan, P., **Roche, C.** 2026; “*Where Cluster Galaxies Reside and How Strongly They Lens*”, *submitted, ApJ*

[†]on a scale of 1-5, where 1 is the highest grade.

Nth Author

1. Lee, W., et al. (incl. **Roche, C.**) 2026; “*TNG-Cluster SIDM: Evolution of massive cluster mergers under the self-interacting dark matter paradigm*”, (in prep)
3. Rose J. C., et al. (incl. **Roche, C.**) 2025; “*The DREAMS Project: A New Suite of 1,024 Simulations to Contextualize the Milky Way and Assess Physics Uncertainties*”, [arXiv:2512.00148](#)
2. Nguyen, T., et al. (incl. **Roche, C.**) 2024; “*How DREAMS are made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds*”, ApJ, 997, 336, [arXiv:2409.02980](#)
1. Rose J. C., et al. (incl. **Roche, C.**) 2024; “*Introducing the DREAMS Project: DaRk mattEr and Astrophysics with Machine learning and Simulations*”, ApJ, 982, 68, [arXiv:2405.00766](#)

Non-Refereed

1. **Roche, C.** & Garnavich, P. 2020; “*Testing Progenitor Models Using the Late-Time Light Curve of Supernova 1992A*” Res. Notes AAS 4 207, [doi:10.3847/2515-5172/abcb05](#)

TEACHING

Massachusetts Institute of Technology

Average student rating 6.95/7 — highest-rated[‡] across 55 semesters of recorded 8.13/8.14 evaluations.

Teaching Assistant, Experimental Physics II (8.14)	Spring 2026
Teaching Assistant, Experimental Physics I (8.13)	Fall 2024
Teaching Assistant, Experimental Physics I (8.13)	Spring 2024
Teaching Assistant, Experimental Physics I (8.13)	Fall 2023
Teaching Assistant, Experimental Physics I (8.13)	Fall 2022
Instructor - Gaia DR3 Hackathon	June 2022

University College Cork

Teaching Assistant, Introductory Physics II	Spring 2018
Leader, Peer Assisted Learning Scheme	2016 – 2018

PRESENTATIONS

Summary: 2 guest lectures, 13 invited talks, 7 conference contributions.

Guest Lectures

Experimental Physics I – Massachusetts Institute of Technology	Oct 2026
Graduate Mathematical Physics – University of Tübingen	July 2021

Invited (Colloquia/Seminars)

Talk “ <i>Calibrating Strong Lenses for Precision Dark Sector Science</i> ” Harvard University	Dec 2026
Talk “ <i>Are Cluster Galaxies Too Good at Lensing?</i> ” MIT Kavli Institute for Astrophysics and Space Research	Oct 2026
Talk “ <i>Are Cluster Galaxies Too Good at Lensing?</i> ” Stanford University	Oct 2026
Talk “ <i>Illuminating the Dark Sector with Strong Gravitational Lensing</i> ” Stony Brook University	Sep 2026
Talk “ <i>Illuminating the Dark Sector with Strong Gravitational Lensing</i> ” University of Michigan, Dept. of Astronomy	Sep 2026
Talk “ <i>Illuminating the Dark Sector with Strong Gravitational Lensing</i> ” University of Michigan, Dept. of Physics	Sep 2026

[‡]Among teaching staff with a minimum of 3 evaluations.

Talk	<i>“Light Cone Strong Lensing in Cosmological Hydrodynamical Simulations”</i>	Mar 2026
	MIT Kavli Institute for Astrophysics and Space Research	
Talk	<i>“Understanding Dark Matter with Simulated Strong Lensing”</i>	Sep 2025
	Yale University	
Talk	<i>“Dark Matter Self-Interaction via Galaxy Cluster Dynamics”</i>	Feb 2024
	Harvard University	
Talk	<i>“Galaxies Lacking Dark Matter Produced in a Cosmological Simulation”</i>	Apr 2022
	MIT Kavli Institute for Astrophysics and Space Research	
Talk	<i>“Exact parallel waves in general relativity”</i>	June 2021
	Mathematical Physics Colloquium – University of Tübingen	
Talk	<i>“How to Stop a Hurricane: Complexity in Physics”</i>	Oct 2020
	“Blow Your Mind Week” 2020 – University College Cork	
Talk	<i>“Hermiticity of Young Operators”</i>	June 2020
	Young Tableaux Seminar Series – Universität Tübingen	

Contributed

Talk	<i>“The Dark Universe Through the Strong Lens: Cluster- and Galaxy-Scale Probes of the Dark Sector” (dissertation talk, submitted)</i>	Jan 2027
	American Astronomical Society Meeting 249 – Salt Lake City	
Poster	<i>“Are Cluster Galaxies Too Good at Lensing?”</i>	Sep 2026
	Probe Combination for Dark Matter Physics in the Era of Large Surveys – Kavli Institute for Cosmological Physics, University of Chicago	
Poster	<i>“The Line of Sight as a Major Systematic of Cluster Strong Lensing”</i>	July 2026
	Gravitational Lensing at the Turning Point – Sexten Center for Astrophysics	
Talk	<i>“The Observation-Simulation Connection and the Line of Sight in Cluster Strong Lensing”</i>	June 2026
	Strong Gravitational Lensing Workshop – University of Montpellier	
Talk	<i>“Consistent Strong Lensing Predictions from Cosmological Simulations”</i>	Oct 2025
	Scaling up Lensing Workshop – University of Liège	
Poster	<i>“Unveiling Dark Matter: Testing CDM with the First End-to-End Cluster Strong Lensing Simulations”</i>	July 2025
	Tracing cosmic Evolution with Galaxy Clusters V – Sexten Center for Astrophysics	
Talk	<i>“Testing CDM with the First End-to-End Cluster Strong Lensing Simulations”</i>	June 2025
	European Astronomical Society Meeting 2025 – University College Cork	

ADVOCACY AND OUTREACH

MIT GAGA Initiative Co-chair 2022 – 2023 | Organizing Committee 2021 – 2024
The Graduates Advising Graduate Admissions program offers information and mentorship to underrepresented groups in physics, and advises on improvements to the admissions process.

MIT PhysGAAP Program 2021 – 2023
The Physics Graduate Application Assistance Program provides guidance to prospective applicants, and liaises with physics department leadership on issues of accessibility in the graduate application. I have acted as one of the primary organizers and have mentored 5 students through the program.

Astrogazers, MIT 2024 – Present
Student Academic Representative, University College Cork 2017 – 2019

LEADERSHIP AND SERVICE

Admissions Committee Member , MIT Physics Graduate Program	2023 Cycle, 2024 Cycle
<i>Reviewed ~ 300 applications to the MIT physics graduate program.</i>	
Admissions Committee Member , MIT Summer Research Program	2023 Cycle
Volunteer , AAS Meeting 241 Graduate School Fair	2023
Graduate Student Office Manager , MIT Kavli Institute	2022 – 2025
Organizer , MIT Kavli Institute Graduate Student Lunch Series	2022 – 2023
Member , MIT Physics Graduate Student Council	2021 – 2023
Organizing Committee , MIT Graduate Student Union	2021 – 2023
Astronomy Officer , UCC Physics and Astronomy Society	2017 – 2019

EDUCATIONAL RESOURCES

These are some of the educational resources and tools I have produced to support my peers, my students or the community at large.	
figinvert	https://github.com/CianMRoche/figinvert
A lightweight command line tool based in python that inverts vector and raster graphics for dark mode talks. In the default mode, mimics overleaf’s PDF inversion.	
Caustica	https://cianmroche.github.io/assets/caustica/
An educational and intuition-building online strong lensing simulator, also capable of producing high-quality lensing media such as video and vector graphics.	
χ by Eye	https://cianmroche.github.io/assets/chi_by_eye/
A learning tool in a geoguessr-like format for training students to recognize believable or unrealistic statistics in a plot by eye. Sandbox mode is particularly useful for gaining hands-on intuition.	
AstroWiki	https://publish.obsidian.md/astrowiki/
I created and now co-maintain a set of living notes covering most of graduate level astrophysics, specifically designed to aid the next generations of graduate students when studying for the MIT astrophysics oral qualification exam.	